Downside-risk sharing*

Federico Lenzi† Luca Rossi‡

October 12, 2026

Abstract

Risk sharing is typically estimated using models that rely on both tails of the income-growth distribution, yet agents care more about downside risks. We focus on negative income shocks, developing a methodology to estimate channels of downside-risk sharing and extend it to increasingly severe shocks – an issue important to policymakers but overlooked in prior work. We find that federal transfers and taxes absorb a large share of income-growth fluctuations, a result earlier studies understated. Overall, our findings challenge policy arguments based on past research portraying public risk sharing as a relatively weak channel.

JEL classification: E64, H00, H29.

Keywords: Risk sharing, downside risk, federal taxes and transfers.

*We thank Pierfederico Asdrubali, Barry Eichengreen, Daniel Gros, and Anna Rogantini Picco for their discussion of a previous version of this paper, and seminar participants at Banca d'Italia, Banco de España, CEPS-ECFIN, Kansas City Fed Midwest Macro Meeting, EIEF, the 13th International Association for Applied Econometrics Conference, ESCB Research Cluster 2 Workshop, the 3rd International Conference on European Studies, the 20th LAGV International Conference in Public Economics, the XXXII Villa Mondragone International Economic Seminar, the 23rd INFER Annual Conference, and the Fourth Conference on Law and Macroeconomics for useful comments and suggestions. We are also grateful to Luisa Carpinelli, Riccardo Cristadoro, Riccardo Degasperi, Marco Del Negro, Paolo D'Imperio, Andrea Finicelli, Evi Pappa, Javier Pérez, Ivan Petrella, Sergio Santoro, Luigi Federico Signorini, Fabrizio Venditti, and two anonymous referees. The views expressed in this paper are ours and do not necessarily reflect those of Banca d'Italia. All remaining errors are our own.

†Kellogg School of Management, Northwestern University, federico.lenzi@kellogg.northwestern.edu.

‡Banca d'Italia, luca.rossi@bancaditalia.it.

1 Introduction

How much does the existence of a federal government help members of a fiscal union share risks with each other? This question sparked the literature on empirical risk sharing around the time of the European monetary unification (Sala-i-Martin and Sachs, 1992). During this period, economists questioned whether a monetary union could function effectively in the absence of a central government capable of collecting higher taxes from member states experiencing idiosyncratic expansions and transferring resources to those temporarily struggling relative to the broader federation. That is, how much hedging was foregone by not availing of a fiscal risk sharing mechanism.[1]

Insights on risk sharing attainable through the fiscal channel were derived in the literature using the US as a benchmark of a mature federation. The seminal contribution by Asdrubali et al. (1996) (ASY henceforth) proposed an econometric model for the joint estimation of three channels of risk sharing, namely the capital market channel, the credit channel, and the fiscal channel, finding a rather limited role for the latter. Their work marked a turning point, for most of the subsequent literature – including the current one – has built on its foundations, generally corroborating the original results.

We depart from these contributions in two dimensions that were common to all the most recent works on the channels of risk sharing in the US.

First, we concentrate on the extent of risk sharing specifically associated with downside risks, which we call "downside-risk sharing". As we explain below, the channels with which risk affects economic variables usually work through variations in the volatility of *negative* outcomes. While the recent financial, macro-econometric and forecasting literature has clearly recognized the importance of downside risks, the existing works on risk sharing had not yet made the distinction between risk sharing driven by a reduction

[1]Differently from monetary policy, fiscal policy can indeed be targeted or designed to affect specific regions, thereby helping to balance the economic outcomes of union members (Kenen, 1969). Farhi and Werning (2017) provide theoretical support for the usefulness of government intervention within a currency union, whereby the full benefits of risk sharing can be reaped only when such an intervention takes place. In their own words, fiscal integration and financial integration are not perfect substitutes.

in upside risks versus that driven by a mitigation of downside risks.[2] In other words, the effects of risk sharing against negative and positive income shocks are averaged out – without any distinction of the two notions. We introduce a novel methodology to estimate the extent of downside-risk sharing, which is not confined to estimates of fiscal risk sharing, rather it can successfully be applied to any risk sharing channel and at any level of aggregation.

Second, we estimate our downside-risk sharing measures based on household-level data from the University of Michigan Panel Study of Income Dynamics (PSID). The previous literature relied on state-level data regressions that – by construction – completely neglect risk sharing within states.[3] We observe that total risk sharing can be seen as the sum of inter-state and within-state components, and we later show in our application that focusing only on the former provides a partial picture of the capabilities of a given risk sharing channel.[4]

The main finding of this paper is that i) computing risk sharing with downside risks only rather than using both tails of the residual income growth distribution, and ii) using micro data that allow to estimate a broader concept of risk sharing, lead to significantly higher fiscal risk sharing estimates than what the previous literature has found. While ASY and subsequent contributions obtained fiscal risk sharing estimates at approximately

[2] See Giglio et al. (2016), Adrian et al. (2019), Plagborg-Møller et al. (2020), Brownlees and Souza (2021), Adrian et al. (2022), Carriero et al. (2024) and Delle Monache et al. (2024) among others. As recalled by Estrada (2006) and in the references therein, investors associate risk with negative outcomes. When the distribution of stock returns is perfectly symmetric this distinction is irrelevant, yet the concept of standard deviation in the presence of significant skewness is of little use. In such an environment, an investor that aims at minimizing the effects of negative events should estimate the volatility of the left tail only. Markowitz (1959) first provides a way to do this by proposing his concept of semi-variance. In Section 2.3, we adapt his definition to our framework in order to obtain estimates of downside risk in aggregate income measures.

[3] Asdrubali et al. (2020) use *Italian* micro data to explore household risk sharing channels.

[4] As recalled in Asdrubali et al. (2020), "*initial empirical tests of risk-sharing were carried out at the microlevel; [...] however, these studies could only test whether the null hypothesis of full risk-sharing was rejected or not, without being able to identify or measure the economic mechanism at work*". Importantly, the authors highlight that the "*theoretical underpinning*" of the "*macroliterature on interregional/international risk-sharing*" is "*typically a representative-agent extension of the basic framework*". This is clearly stated also in Sørensen and Yosha (1998): "*Consumers within each country are assumed to be identical ex-ante [...] as well as ex-post [...]. Thus, the derivation focuses on risk sharing between countries, ignoring risk sharing within countries*".

15% after two years, our findings in fact indicate a tightly estimated value of 85.5%.[5] Importantly, we show that virtually all risk sharing capabilities are driven by a reduction in the left tail thanks to the federal system of taxes and transfers, while the right tail proves to be remarkably stable after accounting for government intervention. Using state-level data only conceals this result as downside-risk sharing estimates obtained with aggregate data are not significantly different from risk sharing ones.

We provide four additional results. First, we generalize our framework to estimate risk sharing of progressively larger negative shocks, something that we would not be able to achieve with standard risk sharing metrics. We find that the government is able to substantially hedge also *large* negative shocks. Second, we separate the contribution of federal transfers from that of federal taxes and show that the bulk of downside-risk sharing stems from the federal government sending transfers to households. Third, we estimate downside-risk sharing by income subgroups, for policymakers are likely more interested in gauging the fiscal risk sharing effects on the relatively lower-income segments of the population. We find that poorer households are those that enjoy higher income insurance from government stabilizers. Fourth, our baseline measure shows that public risk sharing has increased over time, especially in the period after the Great Financial Crisis.

To the best of our knowledge, we are the first in the risk sharing literature to exclusively and explicitly focus on left tails and gauge the capability of a given mechanism to share *downside* risks.[6]

To fully understand why one should focus on downside risks, it is useful to make a step back and ask why uncertainty matters at all.[7] One of the most important reasons

[5] We use the two-year time interval since the PSID is a biennial survey, which precludes us from providing estimates for non-biennial time intervals.

[6] Dynarski and Gruber (1997) is the only work that acknowledges the importance of income contractions versus expansions, but their paper does not develop a downside-risk sharing methodology. The authors indeed only perform a brief exercise where they estimate their model separately for observations with negative versus positive income growth.

[7] In the following, we use the words "risk" and "uncertainty" interchangeably. This is because the way AVW and we compute risk in the paper is not conceptually different than what done e.g. in Jurado et al. (2015) when they compute macroeconomic uncertainty.

why macroeconomists are concerned with uncertainty is that higher uncertainty leads to lower economic activity, something that is well established in the literature.[8] Bloom (2014) highlights that the channels through which uncertainty displays its effects are mainly two, namely a real options channel and a financial channel.[9] As for the first, Bernanke (1983) lays out what is known as the "bad news principle", whereby he shows that whenever an agent is contemplating to undertake an irreversible investment, only the size and probability of a bad outcome matter for his or her decision to delay the investment rather than paying the sunk cost today. Intuitively, when it becomes more likely that the future payoffs of an investment opportunity will not be worth the investment cost, the agent will have a greater incentive to wait (a higher "option value of waiting") until the uncertain outcome is realized, and invest only if the good state of the world materializes.[10]

The second channel that links uncertainty to economic growth – the financial channel – arises from the fact that increased uncertainty leads investors to ask for higher risk premia, both because of risk aversion and because of heightened default probabilities. This mechanism works through changes in the left tail only. Besides Markowitz (1959), there exists a wealth of literature that either assumes, argues or finds that agents place more weight on losses rather than gains (Roy, 1952; Kahneman and Tversky, 1979; Gul, 1991; Routledge and Zin, 2010; Estrada, 2006). Ang et al. (2006) find the existence of a large downside-risk premium driven by investors who require a higher compensation for holding stocks that co-vary more with the market when the latter is falling; Lettau et al. (2014) provide empirical evidence that only the downside-risk CAPM (as opposed to its traditional counterpart) can explain the cross-sectional dispersion of currency re-

[8] See, for example, Bloom (2009), Christiano et al. (2014), Jurado et al. (2015), Fernández-Villaverde et al. (2015), Baker et al. (2016), Bloom et al. (2018), Carriero et al. (2018), Anzuini et al. (2020), Husted et al. (2020), Anzuini and Rossi (2021).

[9] For a thorough presentation of the fundamentals of real options, see Dixit and Pindyck (1994).

[10] On the other hand, if the probability that the investment will yield the same high returns increases or if the expected magnitude of those favorable future returns rises – leading in both cases to higher upside uncertainty – this will not drive agents to change their behavior, the reason being that if they had known they would find themselves in the good state of the world, they would have invested immediately anyway. More specifically, whenever upside uncertainty increases, agents will regret not to have invested earlier. Since this is not possible, the shape of the right tail simply does not factor into their optimization problem.

turns, and that it can jointly explain the cross section of the returns of a variety of asset classes; Farago and Tédongap (2018) find evidence that – in addition to market returns and volatility – markets price three left-tail-related factors in the cross section of asset returns.

Related literature. Our work is closely related to Athanasoulis and Van Wincoop (2000) and Athanasoulis and Van Wincoop (2001) (AVW henceforth) who develop a novel framework that defines risk sharing as the reduction in the volatility of idiosyncratic shocks after a given risk sharing channel is put in place. This is in contrast with previous contributions – most notably ASY – who compute risk sharing with the covariance between idiosyncratic income growth before and after risk sharing occurs. Crucially, the basic *concept* of risk sharing we use in this paper is the same as AVW, but we differ from them in concentrating on downside risks and developing the first methodology that gauges downside-risk sharing. Section 2.4 discusses the conceptual differences between AVW and ASY.

By using micro data, we get closer to what done by the earliest literature on empirical risk sharing (Cochrane, 1991; Mace, 1991; Nelson, 1994; Hayashi et al., 1996; Dynarski and Gruber, 1997). Asdrubali et al. (2020) note that in order to pin down risk sharing channels at the household level the econometrician necessarily has to control for more confounding factors than what would be required or even possible at the aggregate level, and suggest that this could explain why the bulk of the recent literature opted to use aggregate data. In our paper, we address this difficulty by controlling for more than 30 household- and state-level variables and by estimating the model with a weighted sparse-group Lasso.

As recalled above, the empirical literature on fiscal risk sharing dates back to Sala-i-Martin and Sachs (1992), who perform regressions of either taxes or transfers on disposable income, and find that between one third and one half of an idiosyncratic shock to the average US region is cushioned by the federal government within the same year. Von Hagen (1992) uses GDP instead of personal income and finds low elasticities, thereby

concluding that the US fiscal system does not provide a significant stabilizing function. Obstfeld and Peri (1998) and Poghosyan et al. (2016) obtain similar results using different empirical strategies. Bayoumi and Masson (1995) is an exception, as the paper finds a high degree of stabilization (approximately 30%) granted by the US federal government. However, Mélitz and Zumer (1999) shows that the higher results in this paper as well as Sala-i-Martin and Sachs (1992) have to be attributed to their use of state personal income rather than state GDP.

Sørensen and Yosha (1998) use the method in ASY to estimate risk sharing channels among EMU countries as well as OECD ones; Hepp and Von Hagen (2012) and Hepp and Von Hagen (2013) estimate risk sharing channels among German states, while Burriel et al. (2020) discuss the federal structures of Austria, Belgium, Germany, Spain and the United States and run ASY-like regressions for each of them (except for Belgium whose federal system only consists of three regions).

Milano (2017) extends the approach in ASY and dissects the credit market channel of risk sharing into the private channel, national borrowing, and official credit provided by European institutions. She finds that European countries smooth consumption mainly thanks to counter-cyclical public borrowing, yet after the Great Financial Crisis this concealed heterogeneity between so-called "periphery" and "core" countries, as austerity measures in the former meant that national fiscal policies provided dis-smoothing during the European sovereign debt crisis, which was more than offset by aid from European institutions. The opposite occurred in core countries. Cimadomo et al. (2020) also find that official loans were important drivers of increased risk sharing in the euro area in the aftermath of the crisis. Methodologically, this paper explicitly models time variation in risk sharing by adding interaction terms to the ASY framework.

Ferrari and Rogantini Picco (2023) use a synthetic control method and find that the introduction of the euro led to reduced risk sharing among member countries. Recently, some papers explored heterogeneities in risk sharing in the US (Stempel, 2021) and among

OECD countries (Asdrubali et al., 2023). Cimadomo et al. (2023) extends Asdrubali and Kim (2004) and adopts a full-fledged Bayesian approach to construct time-varying estimates of risk sharing. The authors also propose a test to gauge whether different risk sharing channel are substitutes or complements.

Finally, Alcidi et al. (2023) extends US and EA analyses based on ASY using the most recent data, generally confirming previous results.

2 Methodology

The starting point of our modeling approach is the framework in AVW, which we review in Section 2.1. Section 2.2 proposes two simple but in our opinion necessary modifications to their framework. Then, Section 2.3 advances AVW's methodology by proposing a strategy to assess the extent to which the channel under consideration is capable of reducing semi-variances – namely, downside risk – as defined in Markowitz (1959) rather than standard deviations as in AVW. Finally, Section 2.4 discuss AVW as well as our methodology with a broader perspective. Like other papers, also AVW use state-level data. Here we present the framework for an household-level analysis, yet our methodology can be indistinctly be applied to state-level data with minor modifications.

2.1 AVW – a review

Let $i = 1, \dots, N$ denote household i, ω_t^i a generic income measure, and let $g_{t,t+h}^i = 100 \cdot \left(\frac{\omega_{t+h}^i}{\omega_t^i} - 1 \right)$ be the growth rate of income from period t to $t+h$. While households can insure idiosyncratic risk, they can do little to diversify away aggregate risk. AVW write the difference between state and aggregate growth as the sum of a predictable and an unpredictable component, where the residual corresponds to the growth component which is diversifiable. In our application, we use the equivalent approach of including

time fixed effects. Therefore, the model to be estimated is

$$\begin{aligned} g^{b,i}_{t,t+h} &= \beta'^{b}_{h} \mathbf{x}^{b,i}_{t,h} + \epsilon^{b,i}_{t,t+h}, \\ g^{a,i}_{t,t+h} &= \beta'^{a}_{h} \mathbf{x}^{a,i}_{t,h} + \epsilon^{a,i}_{t,t+h}, \end{aligned} \tag{2.1}$$

where the superscripts b and a refer to variables from regressions whose dependent variable is income measured before and after risk sharing, respectively. $\mathbf{x}^{\cdot,i}_{t,h}$ is a $(K+1) \times 1$ vector of control variables which includes a constant term, fixed effects, dummy and continuous variables, and we assume that $\epsilon^{\cdot,i}_{t,t+h}$ is a forecast error with mean zero and variance $\sigma^{2,\cdot}_{h}$. This system is estimated using non-overlapping intervals of length h, which is necessary in order to obtain independent innovations.[11]

In this setting, $\sigma^{2,b}_{h}$ has to be interpreted as a measure of diversifiable risk. AVW define the amount of risk sharing accomplished by a given channel as the relative reduction in the residual standard deviation of income growth after risk sharing as compared to the one before it. Therefore, their estimates of risk sharing at horizon h are given by

$$RS^{*}_{h} = 100 \cdot \frac{\hat{\sigma}^{b}_{h} - \hat{\sigma}^{a}_{h}}{\hat{\sigma}^{b}_{h}}. \tag{2.2}$$

Crucially, this modeling approach does not require to recover structural parameters: all that is needed is to obtain reliable estimates of the conditional mean. Thus, in principle one can use Ordinary Least Squares (OLS) to estimate equation (2.1) and its variants below. Nevertheless, given the considerations above, we prefer to include in our regressions many predictors and not use OLS, which is known to be prone to over-fitting issues in the presence of many regressors. We instead estimate our model with a sparse-group Lasso (Simon et al., 2013).

Finally, given that our goal is not to test the risk sharing theory but rather to obtain *ag-*

[11] The reliance on non-overlapping intervals is the reason why we need a subscript h in the control variables. Moreover, AVW assume Gaussianity in the error terms and obtain confidence bands with analytic approximations. Here, we instead drop this assumption and perform inference with bootstrapping procedures. Indeed, standard Gaussianity tests on estimated forecast errors reject the null of normality.

gregate measures of *idiosyncratic* risk, we give higher weight to more representative households.[12]

2.2 AVW – modifying the basic framework

In this section, we acknowledge that the way risk sharing is computed in AVW can first be improved over some dimensions, and we therefore first examine those intermediate steps. First, we believe that computing risk sharing with variances rather than standard deviations as in AVW is more appropriate, for variances are additive while standard deviations are not: if we normalize risk reduction by total variation rather than the residual one (as we suggest in the next paragraph) we then necessarily have to recur to variances. We therefore first modify equation (2.2) as follows:

$$RS_h^{**} = 100 \cdot \frac{\hat{\sigma}_h^{2,b} - \hat{\sigma}_h^{2,a}}{\hat{\sigma}_h^{2,b}}. \tag{2.3}$$

Now, by means of a numerical example we argue that computing risk sharing as in equations (2.2) or (2.3) can lead to misleading estimates. Let's first assume that estimating the model in equation (2.1) leads to point estimates $\hat{\sigma}_h^b = 20\%$ and $\hat{\sigma}_h^a = 10\%$ for some h. In this case, equation (2.3) suggests that the channel under consideration is able to hedge $100 \cdot \frac{0.2^2 - 0.1^2}{0.2^2} = 75\%$ of idiosyncratic risk in income growth. In economic terms, this result points to this channel being particularly useful, as the risk reduction is sizeable not only in percentage terms but also in absolute ones (a ten *percentage points* reduction in income growth risk). Now, suppose instead that fitting the same model on the same variables yields estimates $\hat{\sigma}_h^b = 0.5\%$ and $\hat{\sigma}_h^a = 0.25\%$. Also in this case, equation (2.3) would imply the same 75% risk sharing value. *However*, it is clear that risk reduction in absolute terms is now much lower, equal to a mere fourth of a percentage point as compared to ten in

[12] In the following exposition, we aim at being as intuitive as possible and therefore outline how we would calculate risk sharing if we were *not* to weight observations by population or – in the PSID application – by household weights. Weighted OLS regressions (used in models with state-level data) need slightly different computations, which we describe in Appendix A.

the previous scenario. In other words, the latter high risk sharing estimate is of little economic relevance as idiosyncratic risk was already low *before* risk sharing took place. Generally speaking, the same values of both RS_h^* and RS_h^{**} in equations (2.2)-(2.3) are consistent with a whole range of potentially very different values of the amount of risk that agents would like to insure against.

Given all the above, we therefore deem that risk sharing should be computed not as a percentage of the before-risk-sharing *residual standard deviation* as is done in AVW, but as a share of *total variance*. Thus, we further modify the calculation of risk sharing:

$$RS_h = 100 \cdot \frac{\hat{\sigma}_h^{2,b} - \hat{\sigma}_h^{2,a}}{\hat{\sigma}_h^{2,b,tot}} = RS_h^{**} \frac{\hat{\sigma}_h^{2,b}}{\hat{\sigma}_h^{2,b,tot}} < RS_h^{**}, \tag{2.4}$$

where $\hat{\sigma}_h^{2,b,tot}$ is the sample variance of our before-risk-sharing dependent variable at every horizon. Note that this adjustment leads to risk sharing estimates that are mechanically *lower* than those obtained in equation (2.3).

Continuing with the example above, assume that $\hat{\sigma}_h^{b,tot} = 30\%$. In the first case, we would now obtain risk sharing estimates equal to $75 \cdot \left(\frac{0.2}{0.3}\right)^2 \approx 33.3\%$, while in the second case they would drop to $75 \cdot \left(\frac{0.005}{0.3}\right)^2 \approx 0.02\%$, reflecting the lower absolute risk reduction. In the extreme case where $\hat{\sigma}_h^b = \hat{\sigma}_h^{b,tot}$, we would obtain "adjusted" risk sharing values equal to those in equation (2.3). On the other hand, in the opposite scenario where $\hat{\sigma}_h^b \to 0$ and we keep $\hat{\sigma}_h^b = 2\hat{\sigma}_h^a$, our formula would suggest zero risk sharing, while equation (2.3) would always yield (with the example above) a 75% hedging capability.

2.3 Downside-risk sharing

This paper aims at providing estimates of *downside*-risk sharing. In the following we therefore modify the framework above to obtain them. Markowitz (1959) first acknowledges the limitations of standard deviations as a way to compute the underlying risk of stock prices: large positive returns should not contribute to increase risk estimates of a

certain asset. Nevertheless, the mathematical definition of variance inherently neglects signs and treats positive and negative returns in the same way. Therefore, Markowitz (1959) introduced the concept of semi-variance: for a given series $\{r_t\}_{t=1}^T$ of stock returns, he defines conditional returns $r_t^0 = \min(r_t, 0)$ and computes the sample variance of $\{r_t^0\}_{t=1}^T$ as a measure of the downside risk of that asset.

In our paper we apply the same definition, but we do this on the series of *residuals* we obtain from estimating the system in equation (2.1). The underlying reasoning is the same as the one in Markowitz, simply converted to fit in our macro-forecasting setting: we are interested in gauging how much the risk sharing channel we are considering is able to hedge against unexpected left-tail events. Computing risk sharing as in AVW would bias the risk measure we are interested in towards the volatility of unexpected *positive* idiosyncratic income growth events.

Operationally, we then define

$$\begin{aligned}
\hat{\epsilon}_s^{b,i,0} &= \min(\hat{\epsilon}_s^{b,i}, 0), \\
\hat{\epsilon}_s^{a,i,0} &= \min(\hat{\epsilon}_s^{a,i}, 0), \\
\hat{\sigma}_h^{2,\cdot,0} &= \frac{1}{S_h \cdot N - K - 1} \sum_{s=1}^{S_h} \sum_{i=1}^{N} \left(\hat{\epsilon}_s^{\cdot,i,0}\right)^2, \\
g_s^{b,i,0} &= \min(g_s^{b,i} - g_s^b, 0), \\
\hat{\sigma}_h^{2,b,0,tot} &= \frac{1}{S_h \cdot N - K - 1} \sum_{s=1}^{S_h} \sum_{i=1}^{N} \left(g_s^{b,i,0}\right)^2,
\end{aligned} \tag{2.5}$$

where the index s refers to each non-overlapping interval for a given horizon h, S_h is the total number of intervals for that given horizon, and g_s^b is aggregate income growth. One can easily show that the variance of a mean-zero variable (like regression residuals) is the simple sum of semi-variances. Therefore, the larger the difference between the left and righ tail, the less the overall variance will properly approximate the left semi-variance.

We define our baseline downside-risk sharing measure as:[13]

$$DRS_h^0 = 100 \cdot \frac{\hat{\sigma}_h^{2,b,0} - \hat{\sigma}_h^{2,a,0}}{\hat{\sigma}_h^{2,b,0,tot}}. \tag{2.6}$$

A final word of caution is in order: we do not claim that our proposed metric captures welfare improvements for risk-averse households, nor that it necessarily correlates with precautionary savings. Addressing these issues would require complementing our work with both normative and additional positive analyses, which we believe go beyond the scope of this paper. Rather, our objective is to measure downside-risk reduction stemming from the functioning of a given risk-sharing channel. In our application, this is equivalent to estimating the stabilization role of automatic federal transfers and reduced federal tax payments triggered by negative idiosyncratic income shocks.

The simplicity and flexibility of our framework enable us to extend our downside-risk analysis even further. One might argue that households (or states) are also interested in understanding the federal government's capacity to respond to *large* negative shocks. Therefore, we generalize our framework and define downside risk by setting progressively negative thresholds in the conditional forecast errors distribution. Appendix B reports the generalized formulas.

2.4 On the importance of using residual volatility

We develop our methodology by building on AVW, who estimate risk sharing within a different conceptual framework than ASY, in particular by gauging the extent to which a given risk sharing channel reduces the *residual* volatility of income growth. Interestingly, their definition of risk predates later contributions such as Jurado et al. (2015) and Fernández-Villaverde et al. (2015) who define uncertainty as the volatility of the *unex-*

[13] In this case, it could happen that models where the control variables explain little variation could lead to residual *semi*-variance being slightly *higher* than its total counterpart, thereby leading adjusted risk sharing values to be higher than non-adjusted ones.

pected variation of a given variable.[14] As exemplified in AVW, if the correlation between income before and after risk sharing is zero with the two series having the same residual standard deviation, ASY would imply the existence of perfect risk sharing, while AVW would point to no risk sharing at all. Moreover, the framework in AVW does not require to assume a particular process for income, whereas ASY needs to impose that GDP is exogenous.

Empirical work focusing on the estimation of structural parameters has to deal with potentially biased estimates stemming from – among other things – omitted variable bias and contemporaneous causality. ASY abstracts from this issue by assuming idiosyncratic GDP growth to be exogenous; under this assumption, there is no need to control for the predictable variance of income growth. *Nevertheless*, later empirical contributions – see, for example, Mélitz and Zumer (1999) and Asdrubali and Kim (2004) – acknowledge the problems of previous static empirical models and control for potentially relevant variables.[15] As Asdrubali and Kim (2004) state: "*It is somewhat surprising – given the economic structure of the issue at hand – that the literature has so far neglected the use of dynamic and simultaneous econometric models [...] our approach makes clear that a simple static regression of a smoothing channel on [GDP] [...] is subject to a bias, due to the omitted variables [...] and their lagged values, and due to the neglect of the dynamics and interaction among smoothing channels*".

In general, and differently from ASY, the aim of AVW – and our paper – is not to test the theory of risk sharing, but rather to properly estimate ex-ante risk and gauge the extent of its reduction thanks to different risk sharing channels. From AVW: "*ASY do not control for the predictable component of growth rates. It is important to do so if one wants to measure risk sharing, which is about reducing risk exposure rather than smoothing a predictable income path*".

[14] In general, the literature on uncertainty has clarified the importance of distinguishing between predictable variations and the volatility surrounding the unpredictable component of any series. See Jurado et al. (2015) for an in-depth discussion.

[15] ASY themselves have a short paragraph (p. 1103) where they show that in their application results do not change significantly when they control for lagged income growth rates.

In addition to the above considerations – which apply to all risk sharing channels – we argue that controlling for the predictable variation is even more important if one is interested in estimating fiscal risk sharing. As for what concerns the fiscal channel, we argue that *all* the reduction in the variance of income growth that is due to the predictable component is *not* due to risk sharing. Rather, it is a by-product of fiscal redistribution.[16] We explain this statement with a simple example. Suppose the existence of a federation which consists of only two states A and B with the same population which is constant over time. Both states have per capita GDP *levels* equal to $100 in the base period, and it is *known* with certainty that – thanks to fixed investments made in the past – A will grow by 6% during the year, whereas – because of structural inefficiencies in the economy – B will contract by the same amount in the same period, which implies zero growth nationwide and state growth rates equal to idiosyncratic ones. At the end of the period, the federal government steps in by taxing A $2, transferring this amount to B. Therefore, growth rates after government intervention will equal 4% (as opposed to 6%) in A, and -4% (as opposed to -6%) in B. In other words, federal government intervention reduced the variance of income growth. However, we know that states in this federation were not hit by idiosyncratic income shocks (we are assuming a deterministic world). Rather, their different growth rates were due to structural *and therefore predictable* reasons (past investments in one country, inefficiencies in the other). Thus, the federal tax and transfers payments and the consequent reduction in the variance of income growth were due to redistribution as opposed to risk sharing. In general, and as we said in the paper, variance reduction stemming from persistent (structural) differences in growth rates between states is rather due to fiscal redistribution than risk sharing, as those differences are indeed predictable. Any model that is estimated without accounting for this will obtain biased estimates (most likely overestimates) of the risk sharing capabilities of the tax and transfer system. The distinction between predictable and unpredictable variation is also

[16] In this paper, we use the definition of redistribution available for example in Malkin and Wilson (2013).

extremely relevant for policy purposes, for instance in the European Union where the long debate regarding risk sharing, redistribution, and fiscal discipline needs to be properly informed.

3 Data and estimation strategy

We apply the model developed in Section 2 to the fiscal channel of downside-risk sharing in the US, and we exploit household-level data available from the PSID, a longitudinal household survey. To further motivate the importance of estimating risk sharing using micro data, we report a few stylized facts: according to the 2019 PSID wave, average per capita income is representative of incomes at the 67th percentile of the distribution. This means that using aggregate state-level data implicitly provides risk sharing estimates that focus on relatively affluent individuals, who are also those that depend relatively less on the proper functioning of risk sharing channels, since they can usually count on stronger savings buffers. In addition, state-level data conceal the effects both of federal interventions that are typically targeted at lower-income households, and of federal tax payments that affect disproportionately more higher-income ones. In 2019, 50% of all social security benefits were received by households in the bottom 23% of the income distribution, and the top 11% of income earners paid 50% of all federal taxes. The above descriptive evidence is a useful starting point to explain why as of today the literature has recovered relatively small effects of the fiscal risk sharing channel. The results emerging from our framework challenge these previous findings.

During the years, the PSID included individuals that joined the family and excluded those leaving it. Moreover, starting from 1997 it also started to properly account for the flow of new immigrants. We restrict our attention to the Survey Research Center (SRC) sample which selected 2930 families representing the entire US population. The dataset provides a set of weights which allows researchers to obtain a sample that is represen-

tative of the US society in any given year. More specifically, we weight all observations according to the procedure illustrated in Gouskova et al. (2008). These weights reproduce the US population, compensating both for unequal selection probabilities and differential attrition.

For our purposes, in order to construct the dependent variables to be used in equation 2.1 – recall that to obtain risk sharing estimates within this framework we have to run two regressions – we need total taxable income, federal transfers received and federal taxes paid. The transfers series covers amounts received by the household head and spouse during the year preceding the interview. It includes Temporary Assistance for Needy Families (TANF), Social Security income, Supplemental Security income, veteran pensions, other retirement pensions, Individual Retirement Arrangements (IRA), unemployment compensation, worker compensation, child support income, heating subsidies, Supplemental Nutrition Assistance Program (SNAP), and other welfare-related transfers. Financial assistance received from relatives is excluded.

As for federal income taxes, the PSID does not directly provide them, but we impute them using the PSID measure of total taxable income for the household head and spouse and the historical income-tax brackets and statutory rates available in Tax Foundation (2025) for the income reference year. Filing status is approximated using marital status and the presence of dependents. We assign married filing jointly to married couples, married filing separately to separated but not divorced individuals, and single or head-of-household status to unmarried individuals without or with dependents, respectively.

Our income growth measure before risk sharing is equal to taxable income growth, while the corresponding measure after risk sharing first adds federal transfers and subtracts federal taxes before computing growth rates.[17]

Given the high number of regressors included, we estimate our baseline model with

[17] A low number of individuals enter the job market for the first time, thereby recording an extremely high income growth rate. To avoid our results to be driven by those observations, we censor these values at 600%.

a sparse-group Lasso, which allows the researcher to include many variables and letting the algorithm decide which group of them to maintain and whether to shrink them or not. We include a constant term, the lagged dependent variable, household size, age of the head, education level of the head of household, education level of the wife, number of children, age of the youngest child, interview length, total working hours of head and wife, average hourly salary for head and wife, working days lost for sick leave of head and wife. We also control for the most relevant variables at the state level that were used in AVW, namely the log of per capita income, the growth rate of per capita income, the five-year lagged population growth rate, and the ratio of state government expenditure to GDP. In addition, we include the following set of dummy variables: year, state of residence, year and state of residence interaction, Division, family composition change, sex of the head, marital status of the head, dummy variable for new wife, primary race of the head and wife, employment status of the head and wife, work for pay of head, wife, or both of them, dummy for self-employed head, corporate job for the head, public job for the head, unionized job for the head, mode of interview.

Section 4 compares estimates obtained from PSID data with those based on data from the Bureau of Economic Analysis (BEA). As for the latter, the closest definition of income we can use is per capita personal income net of transfers. We measure state-level federal taxes as the sum of employee and self-employed contributions, and personal current taxes to the federal government. Appendix C describes how we compute federal transfers. State-level regressions include few regressors and we thereby estimate them with Weighted Least Squares, using population weights.

We estimate equation (2.1) on PSID data and compute downside-risk sharing for the 1997-2019 period. The reason why we focus on this time span is twofold. First – as Figure 3.1 shows – federal government intervention in the form of direct or indirect transfers to individuals has substantially increased over time. If we look at a sample starting in 1963 (the starting date for many past studies) one can see that those transfers weighed

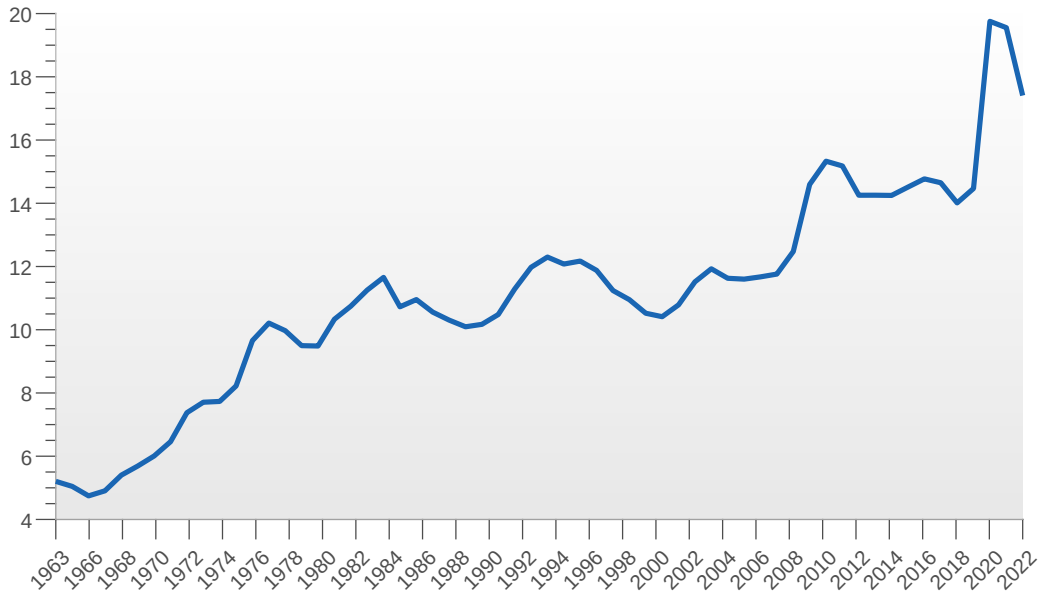


Figure 3.1: Total payments for individuals. The figure shows the evolution of the sum of direct payments for individuals and grants to state and local governments aimed at the personal sector, in percent of GDP. Data are taken from the Historical Tables of the Office of Management and Budget (OMB).

5% of US GDP at the beginning of the sample, rapidly increased during the '70s, fluctuated around 10-12% during the Great Moderation, and then rose in the aftermath of the Great Financial Crisis and especially after the pandemic. Limiting our analysis to 1997-2019 avoids using the earlier sample when the role of the federal government was significantly lower, and ensures our results are not driven by the unprecedented federal aid disbursed to US households especially during 2020 and 2021. Moreover, below we compare results obtained with PSID data with those derived using state-level data from the BEA. Therefore, the second reason why we start our estimation from 1997 is that in this year the BEA replaced SIC codes with NAICS ones, thus by using only post-1997 we ensure accounting consistency in state-level regressions.

Finally, for the household-level estimates, we perform inference using a household-cluster bootstrap similar to the cross-sectional resampling procedure of Kapetanios (2008). We stratify households by their initial-income quartile, based on taxable income in the first year a household appears in the sample, and, in each replication, draw households

with replacement within each quartile, so that each quartile keeps its original number of households in the estimation sample. We repeat this procedure 1,000 times.

Each selected household contributes its complete observed history and its original PSID longitudinal weights. Resampling household histories preserves within-household serial dependence, while stratification preserves the sample allocation across initial-income groups without imposing equal population weights on the four quartiles. Our risk-sharing measures are computed from out-of-fold forecast errors obtained using five-fold cross-validation at the household level. In each bootstrap replication, we hold the estimated forecasting models, selected penalties, and associated out-of-fold predictions fixed. We then recompute the weighted national mean growth rate of each period, which enters the downside benchmark, the weighted second moments of forecast errors and of total income growth, and the resulting risk-sharing measures, including those by income quartile and for each loss threshold.

4 Results

The upper part of Table 1 reports results obtained by estimating fiscal downside-risk sharing with PSID data at the household level. We find that, contrary to many past contributions, fiscal risk sharing cushions a large share of idiosyncratic income fluctuations. After two years, it is able to reduce downside risks by 85.5%, with this share progressively lowering as the horizon increases, until it reaches 60.5% after six years, still a sizable estimate. These values are higher than those found in many past research, strongly questioning the conclusion that the fiscal channel plays a relatively limited role in hedging income shocks. Moreover, the table indicates that the uncertainty around the point estimate is small.

To highlight the importance of using household-level data rather than aggregate data, Table 1 also reports the results of an additional exercise in which we estimate our baseline model using household data aggregated at the state level. Estimates based on state-level

Table 1: Fiscal downside-risk sharing: household- and state-level data from the PSID

			90%		95%	
	Horizon	Point estimate	Lower	Upper	Lower	Upper
Household						
	$h = 2$	85.5	82.0	88.9	81.1	89.6
	$h = 4$	65.0	58.4	71.7	57.1	73.2
	$h = 6$	60.5	56.3	64.8	55.3	65.9
State-level						
	$h = 2$	32.0	23.9	43.2	23.0	45.1
	$h = 4$	38.8	18.0	69.4	16.6	78.5
	$h = 6$	39.0	26.4	53.0	23.0	55.6

The table reports results obtained by estimating our downside-risk sharing model on household-level PSID data as well as PSID data aggregated at the state-level. 90% and 95% empirical confidence bands are reported. Figures are expressed as percentages. Horizons are expressed in years.

data always lie well below those obtained from micro data, suggesting that focusing on inter-state risk sharing only – as is common in the literature – provides at best a partial assessment of *total* risk sharing. Our evidence therefore cautions against treating US states as representative agents. Indeed, results in Table 1 indicate that within-state risk sharing – proxied by the difference between household- and state-level estimates – is as important as inter-state risk sharing.

Figure 4.1 dissects our findings – based on PSID data – by showing which is the contribution of each of the variations we impose on AVW's methodology. In the top-left panel we provide estimates of risk sharing by using the methodology in AVW. Interestingly, while state-level estimates – not shown – are qualitatively similar to what found in AVW, when exploiting the heterogeneity of micro data we find significantly higher estimates, above 35% in the first four years. The top-right panel reports results obtained by using variances rather than standard deviations. This simple difference alone is responsible for about 20 p.p. increase in risk sharing estimates. The bottom-left panel shows that – as explained in Section 2.2 – normalizing risk reduction by total variance rather than residual variance leads to a mechanical reduction in risk sharing estimates. Finally, the compari-

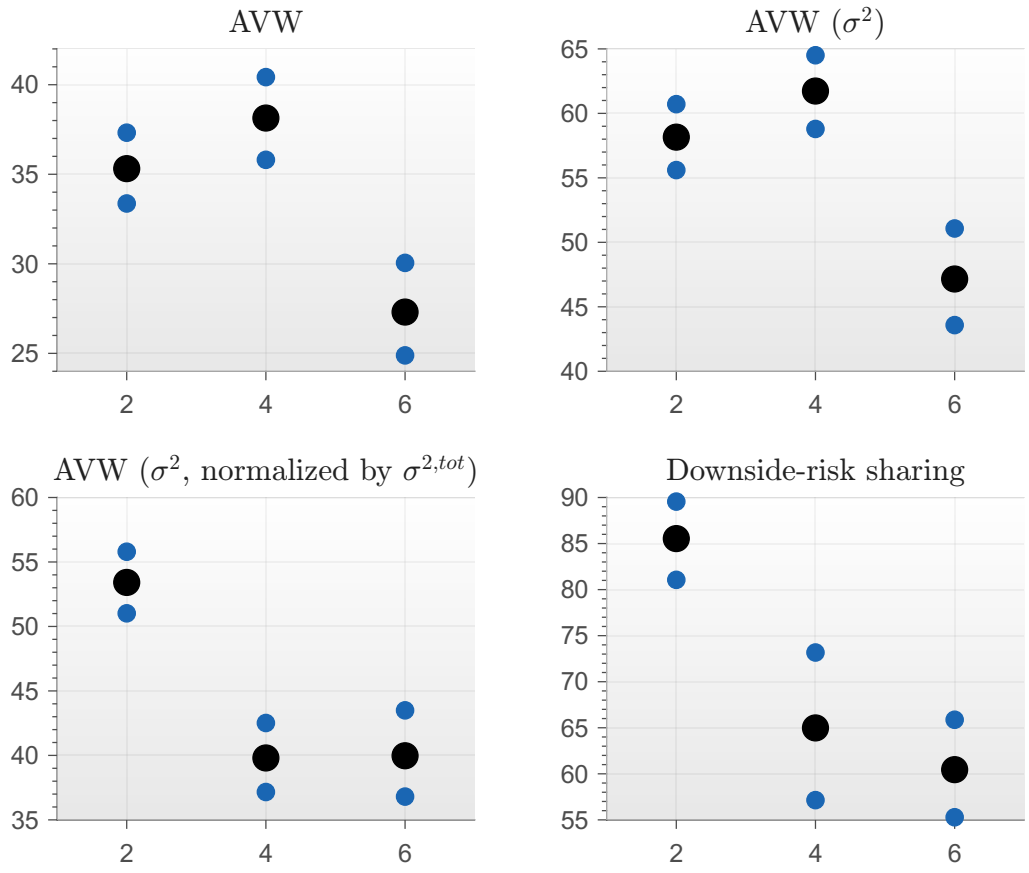


Figure 4.1: From AVW to our baseline. The figure provides estimates obtained by computing risk sharing with different metrics, as explained in Section 2. The x-axis represents the horizon (in years); the y-axis represents the risk sharing estimate (in percent). Black circles are our baseline estimates, taken as the median of the distribution; blue circles are 95% confidence intervals.

son between the bottom-left and bottom-right panels shows the main empirical contribution of this paper: when focusing on downside risks – as in equation (2.6) – as opposed to the whole distribution – as in equation (2.4) – risk sharing estimates starkly rise at all horizons: after two years, risk sharing a-la AVW (but with our modifications) estimates that the federal government provides a 53.4% risk reduction, whereas our downside-risk sharing metric suggests 85.5%. After four (six) years, the former points to 39.8% (40.0%) whereas the latter to a markedly higher 65.0% (60.5%).

Table 2 reports the results we obtain by estimating equations (2.4) and (2.6) on state-level data – described in Section 3 – from the BEA. First, note that estimates based on BEA

Table 2: Fiscal risk sharing and downside-risk sharing: state-level data from the BEA

			90%		95%	
	Horizon	Point estimate	Lower	Upper	Lower	Upper
Risk sharing						
	$h = 2$	20.7	15.0	25.4	14.0	26.1
	$h = 4$	17.9	12.5	23.1	11.5	24.0
	$h = 6$	21.3	15.7	26.9	14.8	27.8
Downside-risk sharing						
	$h = 2$	22.4	17.6	26.2	16.7	26.8
	$h = 4$	17.5	11.0	24.2	9.7	25.5
	$h = 6$	23.4	16.1	31.4	14.9	33.0

The table reports results obtained by estimating fiscal risk sharing and downside-risk sharing as in equations (2.4) and (2.6) using BEA state-level data. 90% and 95% empirical confidence bands are reported. Figures are expressed as percentages. Horizons are expressed in years.

data are similar to those obtained by previous papers. Moreover, aggregating PSID data at the state level (as in Table 1) leads to estimates that are closer to those we obtain with BEA data in Table 2. More interestingly however, Table 2 shows that differences between risk sharing and downside-risk sharing at the state-level are small and not significantly different between them. Therefore, relying solely on state-level data not only understates total risk sharing, but also obscures the significance of downside-risk sharing, making the two measures appear artificially similar.

Figure 4.2 presents residuals from models estimated using both household-level data from the PSID and state-level data from the BEA. Notably, the top panel reveals that for shorter horizons, government intervention effectively mitigates a substantial portion of the left tail, as the left bound of the yellow distribution is significantly closer to zero than that of the blue distribution – particularly at the two-year horizon. However, the right tail remains remarkably stable across all cases. This leads to a crucial implication for fiscal risk sharing in the US: virtually *all* non-zero risk-sharing estimates derived from standard metrics using household-level data are driven solely by the compression of the left tail.

In other words, incorporating the right tail in the computation effectively pushes risk-sharing estimates toward zero. Interestingly, the bottom panel shows that this asymmetry does not appear in state-level data, where reductions in the left and right tails are more balanced.

Finally, Figure 4.3 shows the results we obtain by estimating our model over eight-year rolling windows. Interestingly, when we look at risk sharing estimates that use both tails of the income shocks distribution – namely, the first three panels – we find relatively smaller changes in the insurance role of the US federal fiscal system over time as compared to the steeper upward trend in *downside*-risk sharing estimates (bottom-right panel) starting after the Great Financial Crisis. This is consistent with what we showed in Figure 3.1, namely that federal transfers to individuals markedly increased after 2009.

4.1 Downside-risk sharing from transfers and taxes

Table 3 presents the results obtained after disentangling the effects of federal transfers from those of federal taxes, indicating that most fiscal risk sharing arises from the allocation of federal transfers to households. For the two-year horizon we obtain that federal transfers alone hedge 72.5% of negative income shocks, as opposed to 17.5% for federal taxes. Interestingly, when we sum risk sharing from federal transfers and taxes we closely obtain our baseline estimates shown in Table 1. This reassuring result does not arise mechanically from our methodology. Overall, more than three-quarters of downside-risk sharing can be attributed to federal transfers.

4.2 Downside-risk sharing by income quartiles

Thanks to the availability of micro data, we can potentially explore a wide range of heterogeneities. An interesting one is that based on taxable income differences. In a sense, it is not clear a priori that poorer households should benefit more from risk sharing within

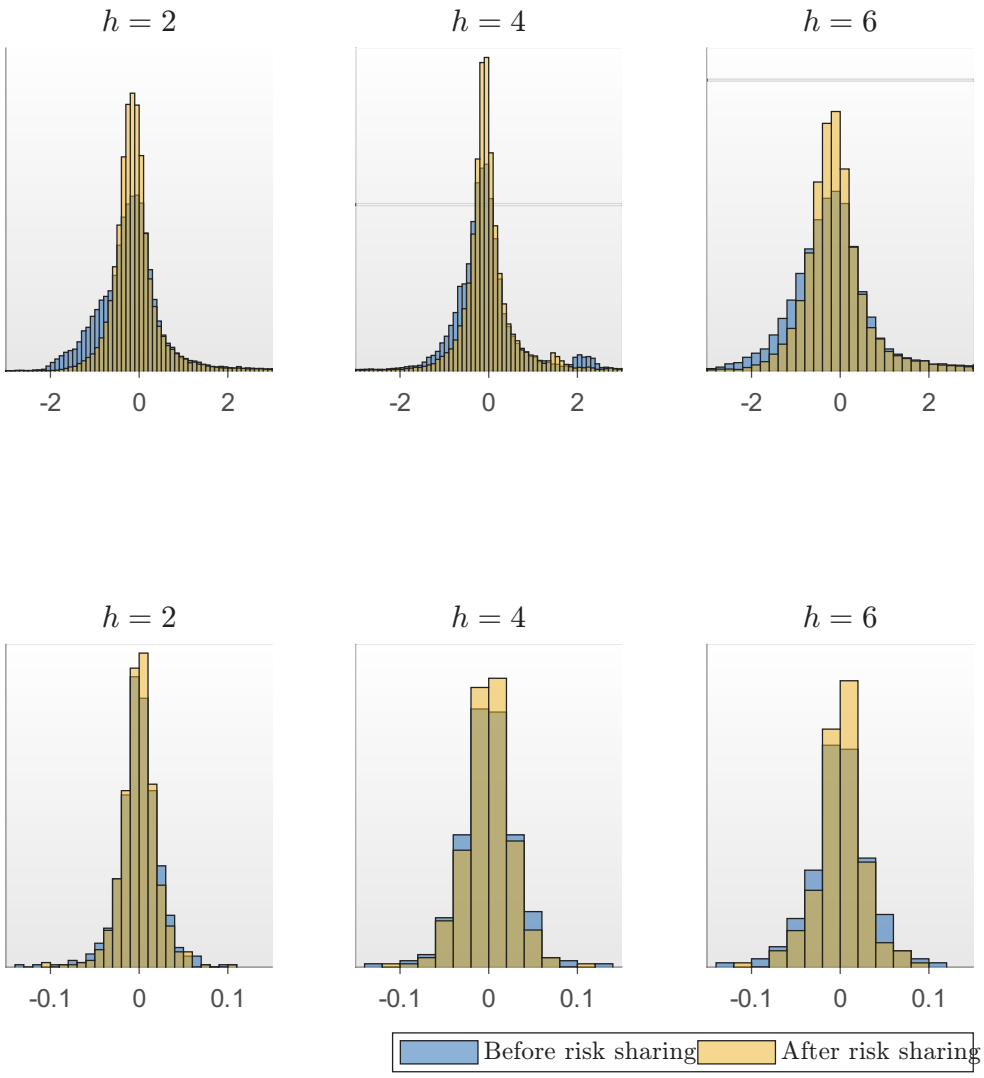


Figure 4.2: Histograms of residuals. The top panels display histograms of residuals estimated from the model using household-level data. Each panel corresponds to a different forecasting horizon. Light blue histograms represent residuals from the model using pre-government-intervention income, while yellow histograms correspond to residuals from the model using post-government-intervention income. The bottom panels present histograms of residuals estimated from the model using state-level data from the BEA.

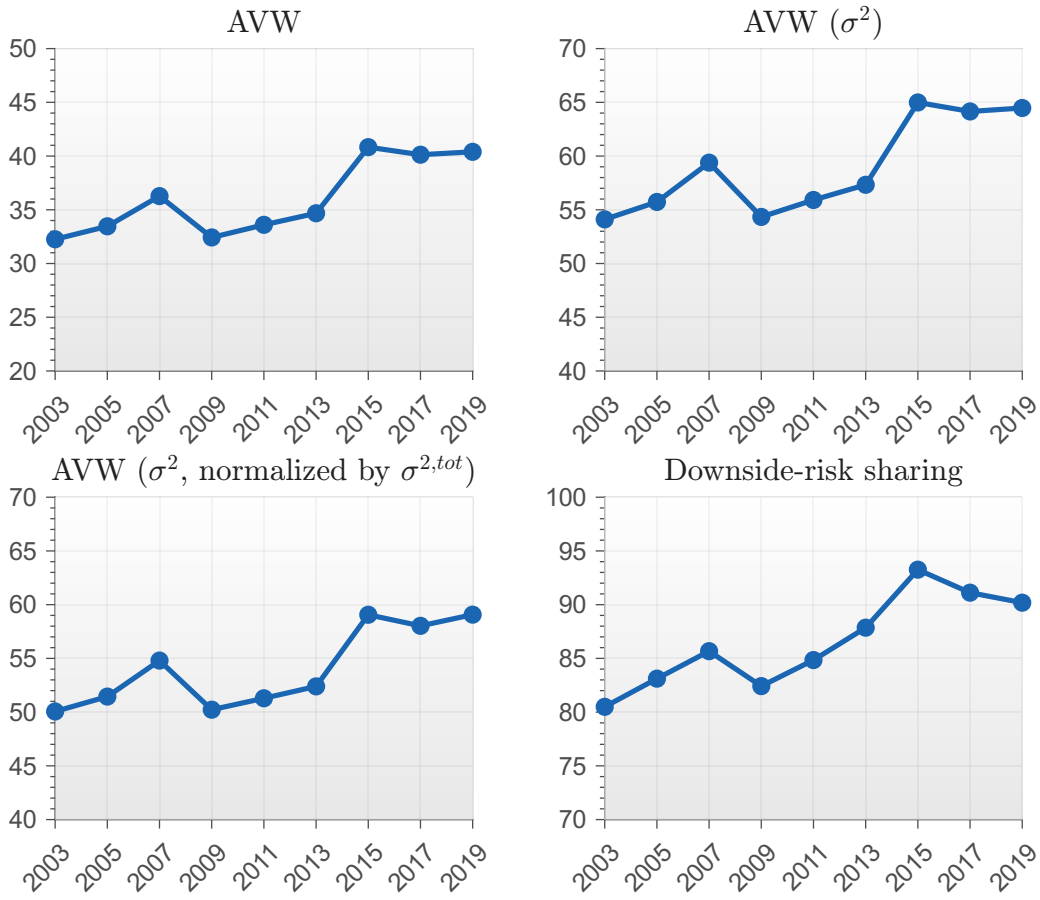


Figure 4.3: Rolling windows. Each panel shows risk sharing estimates obtained through different metrics explained in Section 2. The time series are obtained through eight-year rolling regressions, and refer to risk sharing at the two-year horizon.

Table 3: Fiscal downside-risk sharing: separating the contribution of transfers and taxes

			90%		95%	
	Horizon	Point estimate	Lower	Upper	Lower	Upper
Transfers						
	$h = 2$	72.5	68.3	76.3	67.7	77.2
	$h = 4$	56.3	49.0	63.7	47.6	65.2
	$h = 6$	47.6	42.9	52.3	42.2	53.5
Taxes						
	$h = 2$	17.5	13.8	21.2	13.4	22.0
	$h = 4$	9.3	6.8	12.1	6.5	12.6
	$h = 6$	13.4	10.1	17.1	9.5	17.7

The table reports results obtained by estimating our downside-risk sharing model separately using only transfers and only taxes, on household-level PSID data. 90% and 95% empirical confidence bands are reported. Figures are expressed as percentages. Horizons are expressed in years.

a fiscal federation: on the one hand – and as we showed in Section 1 – the government undoubtedly disburses the majority of social security benefits to low-income households. Nevertheless, richer households are those that pay the largest share of federal income taxes: when they are hit by idiosyncratic negative income shocks, they suddenly start to pay less taxes.

Figure 4.4 reports risk sharing estimates separately by taxable income quartiles and by transfers and taxes. We find clear evidence that lower-income households benefit disproportionately more than higher-income ones: for those in the first quartile, federal transfers cushion all the loss they suffer during the first four years, and 70.1% after six years. On the other hand, federal transfers play a decreasingly important role for higher-income households, especially at longer horizons: for those in the fourth quartile transfers hedge 51.7% of downside risk after two years, but the share drops to 18.7% and 25.6% at the four- and six-year horizons. The finding that upper bands exceed 100% for the first quartile of the distribution is consistent with the fact that a sizeable share of the US population receives more than half of their income from federal transfers (US Department of Health

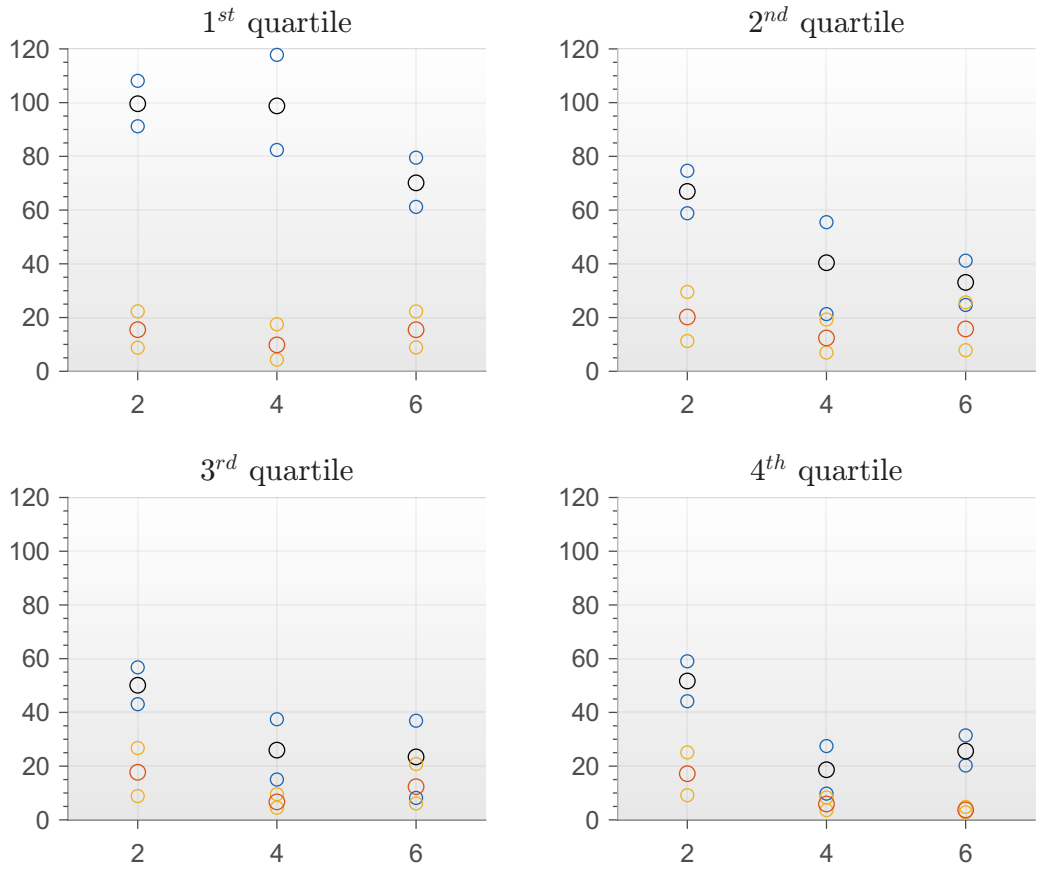


Figure 4.4: Downside-risk sharing by income quartiles. The figure shows downside-risk sharing estimates obtained by running regressions separately for each quartile of the taxable income distribution available in the PSID. The x-axis represents the horizon (in years); the y-axis represents the risk sharing estimate (in percent). Black (orange) circles refer to point-estimates for downside-risk sharing stemming from transfers (taxes); blue (yellow) circles are 95% confidence intervals for transfer (taxes)-related estimates, obtained with the bootstrap procedure described in Kapetanios (2008).

and Human Services, 2021). According to figures reported therein, this share (the dependency rate) equals on average about 4.5% of the US population within our sample. If one assumes that transfer dependency is concentrated entirely within the lowest income quartile, these individuals would account for roughly 18% of that quartile.

The role of taxes is significantly lower and comparatively more stable across quartiles, but also horizons: on average, taxes hedge between 9% and 16% of downside risks across income quartiles.

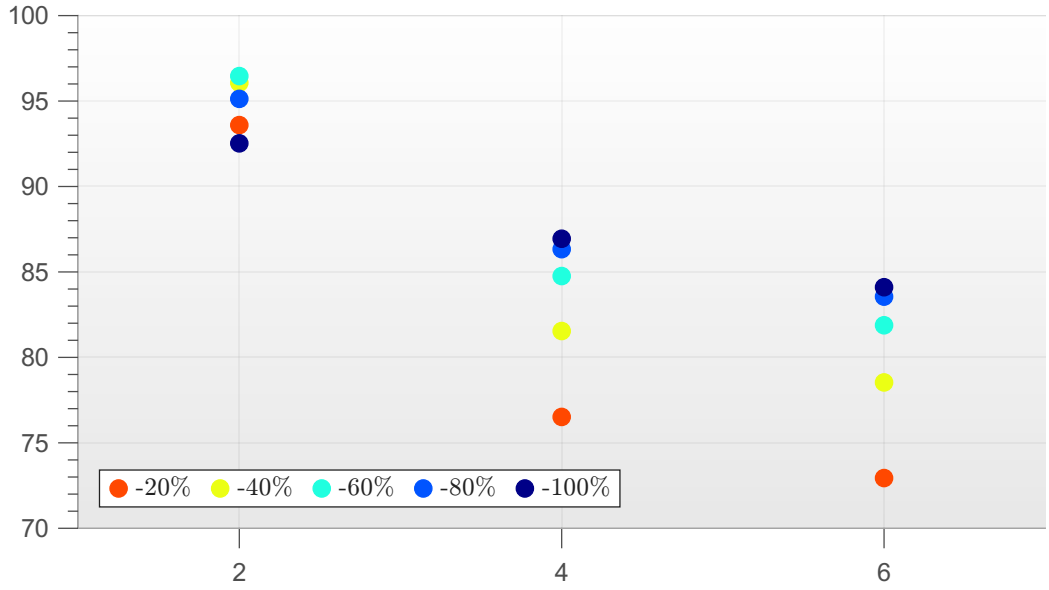


Figure 4.5: Downside-risk sharing with different thresholds. The figure plots downside-risk sharing estimates obtained by setting negative and progressively lower values for c in equation (B.1).

4.3 Downside-risk sharing for larger shocks

Figure 4.5 plots the results obtained from the formulas developed in Appendix B, using a grid of different thresholds c. The figure shows that, as c increases in absolute value, downside-risk sharing tends, if anything, to rise. This suggests that households experiencing severe idiosyncratic shocks can still expect a substantial reduction in income growth variation, at least comparable to that observed among households facing relatively milder shocks.

5 Sensitivity analysis

In addition to the above analyses, in order to gauge the robustness of our results we estimate six further models. First, we deflate income growth with the aggregate PCE deflator. Second, we divide income by household size. Third, we divide income by the square root of household size. Fourth, we get rid of state and division fixed effects. Fifth, we estimate our model with OLS rather than Lasso, and finally we use Ridge regressions. Table

4 shows that the results we obtain are similar to our baseline specification. If anything, using OLS leads to larger risk sharing estimates.

6 Concluding remarks

In this paper, we develop a methodology that allows to estimate downside-risk sharing. Building on the framework in Athanasoulis and Van Wincoop (2001) and adopting a simple definition of (stocks') downside risk from Markowitz (1959), we adapt the latter to our setting to measure downside-risk sharing. We apply our methodology to PSID data to gauge the extent to which the federal government of the United States is able to insure households against idiosyncratic income uncertainty, and we re-evaluate the importance of the fiscal channel by showing that previous estimates vastly understate its capabilities. This is primarily due to two factors. First, using state-level data – as done in most papers – understates total risk sharing because of the implicit reliance on a state-specific representative household to which all income is concentrated. This approach neglects within-state risk sharing, which, in our application, we show to be larger than inter-state risk sharing. Second, computing risk sharing by implicitly using both tails of the income growth distribution also compresses risk sharing estimates. Indeed, we show that virtually all the risk reduction achieved through federal transfers and taxes results in lower volatility of the left tail, while the right tail remains remarkably stable. Interestingly, this phenomenon is not visible when estimating our model with state-level data.

Our work challenges the foundations of many policy debates that – based on past research – assert that the existence of a federal government in the US provides relatively little insurance from idiosyncratic income shocks. In fact, the fiscal risk-sharing channel plays an increasingly important stabilizing role for US households' incomes, especially for those at the lower end.

Table 4: Robustness checks

			90%		95%	
	Horizon	Point estimate	Lower	Upper	Lower	Upper
PCE deflator						
	$h = 2$	86.9	82.9	90.6	82.3	91.5
	$h = 4$	65.7	59.1	72.4	58.1	74.0
	$h = 6$	62.0	57.5	66.3	56.6	67.2
Dividing by HS						
	$h = 2$	76.4	73.1	79.7	72.4	80.4
	$h = 4$	56.3	50.6	62.5	50.0	63.6
	$h = 6$	50.9	46.7	54.8	45.8	55.6
Dividing by $\sqrt{HS}$						
	$h = 2$	83.6	80.2	87.1	79.3	87.7
	$h = 4$	63.0	56.4	69.4	55.3	70.9
	$h = 6$	59.2	54.6	63.8	54.1	64.7
No state or division FE						
	$h = 2$	85.6	82.0	88.9	81.1	89.6
	$h = 4$	65.0	58.4	71.7	57.1	73.2
	$h = 6$	60.2	56.0	64.7	55.1	65.4
OLS estimation						
	$h = 2$	94.7	89.8	99.0	88.8	100.0
	$h = 4$	83.5	74.5	99.6	72.5	94.9
	$h = 6$	80.0	68.8	94.1	66.8	97.1
Ridge estimation						
	$h = 2$	88.3	84.5	92.0	83.5	92.6
	$h = 4$	69.1	62.6	76.0	61.4	76.9
	$h = 6$	67.0	62.5	72.1	61.6	72.9

The table reports results for six robustness checks for each biennial horizon. HS is household size. Numbers (in percent) refer to downside-risk sharing as in equation (2.6).

A Appendix – weighted estimates

Here, we show that estimating our model with weighted regressions requires a slight adjustment to the non-weighted formulas, and therefore first recall some relevant basic facts. For ease of notation, define $y^i_{t,h} = g^i_{t,t+h} - g^{US}_{t,t+h}$, $\mathbf{z}^i_{t,h} = \mathbf{x}^i_{t,h} - \mathbf{x}^{US}_{t,h}$, and $u^i_{t,h} = \epsilon^i_{t,t+h}$. We can therefore rewrite equation (2.1) as:

$$\mathbf{y}_h = \mathbf{Z}_h \boldsymbol{\beta}_h + \mathbf{u}_h, \tag{A.1}$$

where we have stacked all $j = 1, \ldots, S_h \cdot N$ observations and used matrix notation. We then define $\mathbf{W}_h$ to be our (diagonal) weighting matrix, with diagonal elements $w_{jj,h}$ normalized so as their sum equals the total number of observations. We also define $\mathbf{P}_h = \text{chol}(\mathbf{W}_h)$ the matrix obtained by applying the Cholesky decomposition on $\mathbf{W}_h$, which in our case simply yields a diagonal matrix with each element in $\mathbf{P}_h$ being equal to the square root of the corresponding ones in $\mathbf{W}_h$. The weighted version of our model is then performed by estimating the following transformed regression:

$$\mathbf{y}^*_h = \mathbf{Z}^*_h \boldsymbol{\beta}_h + \mathbf{u}^*_h, \tag{A.2}$$

where $\mathbf{y}^*_h = P_h \mathbf{y}_h$, $\mathbf{Z}^*_h = P_h \mathbf{Z}_h$, and $\mathbf{u}^*_h = P_h \mathbf{u}_h$. Morever, define $\hat{\mathbf{u}}_h = \mathbf{y}_h - \mathbf{Z}_h \hat{\boldsymbol{\beta}}^*_h$, with $\hat{\boldsymbol{\beta}}^*_h$ being the weighted estimator.

Because of the fact that also the intercept gets weighted in the transformed regression, only *weighted* residuals $\mathbf{W}_h \hat{\mathbf{u}}_h = \mathbf{P}_h \hat{\mathbf{u}}^*_h$ have a mean equal to zero. This implies that computing conditional residuals as in equation (2.5) or even with the general threshold in equation (B.1) could be misleading.

Therefore, our baseline specification modifies the above equations and define condi-

tional forecast errors in the following way:

$$\hat{u}_{j,h}^{*,c} = \begin{cases} \hat{u}_{j,h}^{*} & \text{if} \quad w_{jj,h} \cdot \hat{u}_{j,h} < c, \\ 0 & \text{otherwise,} \end{cases} \tag{A.3}$$

where c is again a general threshold. Finally, we compute residual semi-variances as

$$\hat{\sigma}_h^{2,*,\cdot,c} = \frac{(\hat{\mathbf{u}}_h^{*,c})' \hat{\mathbf{u}}_h^{*,c}}{S_h \cdot N - K - 1}. \tag{A.4}$$

B Appendix – hedging against larger shocks

Equation (2.5) now becomes:

$$\begin{aligned}
\hat{\epsilon}_s^{b,i,c} &= \min(\hat{\epsilon}_s^{b,i}, c), \\
\hat{\epsilon}_s^{a,i,c} &= \begin{cases} \min(\hat{\epsilon}_s^{a,i}, 0) & \text{if} \quad \hat{\epsilon}_s^{b,i} < c, \\ c & \text{if} \quad \hat{\epsilon}_s^{b,i} \geq c, \end{cases} \\
\hat{\sigma}_h^{2,b,c} &= \left(|c| + \sqrt{\frac{1}{S_h \cdot N - K - 1} \sum_{s=1}^{S_h} \sum_{i=1}^{N} \left(\hat{\epsilon}_s^{b,i,c} \right)^2} \right)^2, \\
\hat{\sigma}_h^{2,a,c} &= \frac{1}{S_h \cdot N - K - 1} \sum_{s=1}^{S_h} \sum_{i=1}^{N} \left(\hat{\epsilon}_s^{a,i,c} \right)^2, \\
g_s^{b,i,c} &= \min(g_s^{b,i} - g_s^{b}, c), \\
\hat{\sigma}_h^{2,b,c,tot} &= \left(|c| + \sqrt{\frac{1}{S_h \cdot N - K - 1} \sum_{s=1}^{S_h} \sum_{i=1}^{N} \left(g_s^{b,i,c} \right)^2} \right)^2,
\end{aligned} \tag{B.1}$$

where c defines the minimum income growth loss for which we want to compute downside-risk sharing. Note that when $c = 0$ the system in (B.1) reduces to that in (2.5).

Downside-risk sharing of larger shocks is thus computed as:

$$DRS_h^c = 100 \cdot \frac{\hat{\sigma}_h^{2,b,c} - \hat{\sigma}_h^{2,a,c}}{\hat{\sigma}_h^{2,b,c,tot}}. \tag{B.2}$$

C Appendix – federal transfers

The main source we use to obtain state-level federal transfers data is the BEA. More specifically, we sum up retirement and disability insurance benefits, Medicare benefits, military medical insurance benefits, income maintenance benefits, veterans' benefits (excluding the category "other assistance to veterans"), education and training assistance, other transfer receipts of individuals from governments, current transfer receipts of nonprofit institutions from the federal government.

Public assistance medical care benefits are a large component and almost exclusively consist of Medicaid and the Children's Health Insurance Program (CHIP), which are administered by states and jointly funded by the federal government and the states. The amount of federal funding for Medicaid is determined by the Federal Medical Assistance Percentage (FMAP), which is based on a formula that provides higher reimbursement to states with lower per capita incomes relative to the national average (and viceversa). FMAPs have been defined in the Social Security Act Amendments of 1965, which created Medicaid by adding Title XIX to the Social Security Act. They grant a statutory minimum of 50 percent and a maximum of 83 percent. The amount of federal funding for CHIP is determined by the enhanced FMAPs (E-FMAP), which are higher than standard FMAPs, although in this case the federal funding share cannot exceed 85 percent. Therefore, we adopt a conservative approach and include only 50% of public assistance medical care benefits.

Unemployment insurance compensation also is jointly funded by federal and state governments, but the BEA only provides aggregate benefits figures. For this reason, many

studies simply include them in their entirety, thereby inducing an upward bias in the actual amount of *federal* unemployment insurance. As pointed out in Alcidi and Thirion (2017), federal funding for unemployment compensation is low during normal times. Nevertheless, it increases both during high state-specific unemployment spells through the permanent Extended Benefits program, and through discretionary federal Emergency Unemployment Compensation schemes that have historically been approved when large symmetric shocks hit the US economy.

In general, it is therefore clear that using BEA's all-government unemployment benefits is unsatisfactory, and we thus depart from all previous contributions by using US Department of Labor state-level data for the federal share of unemployment insurance benefits paid at all levels, namely regular, extended, and discretionary ones.[18]

[18]Past discretionary programmes have been funded by the federal government only.

References

Adrian, T., Boyarchenko, N., Giannone, D., 2019. Vulnerable growth. American Economic Review 109, 1263–1289.

Adrian, T., Grinberg, F., Liang, N., Malik, S., Yu, J., 2022. The term structure of growth-at-risk. American Economic Journal: Macroeconomics 14, 283–323.

Alcidi, C., D'Imperio, P., Thirion, G., 2023. Risk-sharing and consumption-smoothing patterns in the US and the Euro Area: a comprehensive comparison. Structural Change and Economic Dynamics 64, 58–69.

Alcidi, C., Thirion, G., 2017. Fiscal risk sharing and resilience to shocks: lessons for the Euro Area from the US. CEPS Working Document 7.

Ang, A., Chen, J., Xing, Y., 2006. Downside risk. The Review of Financial Studies 19, 1191–1239.

Anzuini, R., Rossi, L., 2021. Fiscal policy in the US: a new measure of uncertainty and its effects on the American economy. Empirical Economics 61, 2613–2634.

Anzuini, R., Rossi, L., Tommasino, P., 2020. Fiscal policy uncertainty and the business cycle: time series evidence from Italy. Journal of Macroeconomics 65.

Asdrubali, P., Kim, S., 2004. Dynamic risk sharing in the United States and Europe. Journal of Monetary Economics 51, 809–836.

Asdrubali, P., Kim, S., Pericoli, F., Poncela, P., 2023. Risk sharing channels in OECD countries: a heterogeneous panel VAR approach. Journal of International Money and Finance 131.

Asdrubali, P., Sørensen, B.E., Yosha, O., 1996. Channels of interstate risk sharing: United States 1963-1990. The Quarterly Journal of Economics 111, 1081–1110.

Asdrubali, P., Tedeschi, S., Ventura, L., 2020. Household risk-sharing channels. Quantitative Economics 11, 1759–7331.

Athanasoulis, S.G., Van Wincoop, E., 2000. Growth uncertainty and risk sharing. Journal of Monetary Economics 45, 477–505.

Athanasoulis, S.G., Van Wincoop, E., 2001. Risk sharing within the United States: what do financial markets and fiscal federalism accomplish? The Review of Economics and Statistics 83, 688–698.

Baker, S., Bloom, N., Davis, S., 2016. Measuring economic policy uncertainty. The Quarterly Journal of Economics 131, 1593–1636.

Bayoumi, T., Masson, P.R., 1995. Fiscal flows in the United States and Canada: lessons for monetary union in Europe. European Economic Review 39, 253–274.

Bernanke, B., 1983. Irreversibility, Uncertainty, and Cyclical Investment. The Quarterly Journal of Economics 98, 85–106.

Bloom, N., 2009. The impact of uncertainty shocks. Econometrica 77, 623–685.

Bloom, N., 2014. Fluctuations in Uncertainty. Journal of Economic Perspectives 28, 153–176.

Bloom, N., Floetotto, M., Jaimovich, N., Saporta-Eksten, I., Terry, S., 2018. Really uncertain business cycles. Econometrica 86, 1031–1065.

Brownlees, C., Souza, A.B., 2021. Backtesting global growth-at-risk. Journal of Monetary Economics 118, 312–330.

Burriel, P., Chronis, P., Freier, M., Hauptmeier, S., Reiss, L., Stegarescu, D., Van Parys, S., 2020. A fiscal capacity for the Euro Area: lessons from existing fiscal-federal systems. Banco de España Occasional Paper 2009.

Carriero, A., Clark, T., Marcellino, M., 2018. Measuring uncertainty and its impact on the economy. The Review of Economics and Statistics 100, 799–815.

Carriero, A., Clark, T.E., Marcellino, M., 2024. Capturing macroeconomic tail risks with Bayesian Vector Autoregressions. Journal of Money, Credit and Banking 56, 1099–1127.

Christiano, M., Motto, R., Rostagno, M., 2014. Risk Shocks. American Economic Review 104, 27–65.

Cimadomo, J., Ciminelli, G., Furtuna, O., Giuliodori, M., 2020. Private and public risk sharing in the euro area. European Economic Review 121.

Cimadomo, J., Giuliodori, M., Lengyel, A., Mumtaz, H., 2023. Changing patterns of risk-sharing channels in the United States and the euro area. European Central Bank Working Papers .

Cochrane, J.H., 1991. A simple test of consumption insurance. Journal of Political Economy 99, 957–976.

Delle Monache, D., De Polis, A., Petrella, I., 2024. Modeling and forecasting macroeconomic downside risk. Journal of Business and Economic Statistics 42, 1010–1025.

Dixit, A., Pindyck, R., 1994. Investment under uncertainty. Princeton: Princeton University Press .

Dynarski, S., Gruber, J., 1997. Can families smooth variable earnings? Brookings Papers on Economic Activity 28, 229–303.

Estrada, J., 2006. Downside risk in practice. Journal of Applied Corporate Finance 18, 117–125.

Farago, A., Tédongap, R., 2018. Downside risks and the cross-section of asset returns. Journal of Financial Economics 129, 69–86.

Farhi, E., Werning, I., 2017. Fiscal unions. American Economic Review 107, 3788–3834.

Fernández-Villaverde, J., Guerrón-Quintana, P., Kuester, K., Rubio-Ramírez, J., 2015. Fiscal volatility shocks and economic activity. American Economic Review 105, 3352–3384.

Ferrari, A., Rogantini Picco, A., 2023. Risk sharing and the adoption of the euro. Journal of International Economics 141.

Giglio, S., Kelly, B., Pruitt, S., 2016. Systemic risk and the macroeconomy: an empirical evaluation. Journal of Financial Economics 119, 457–471.

Gouskova, E., Heeringa, S., McGonagle, K., Schoeni, R., Stafford, F., 2008. Panel Study of Income Dynamics revised longitudinal weights 1993-2005. Survey Research Center - Institue for Social Research, University of Michigan Technical Series Paper .

Gul, F., 1991. A theory of disappointment aversion. Econometrica 59, 667–686.

Hayashi, F., Altonji, J., Kotlikoff, L., 1996. Risk-sharing between and within families. Econometrica 64, 261–294.

Hepp, R., Von Hagen, J., 2012. Fiscal federalism in Germany: stabilization and redistribution before and after unification. Publius: The Journal of Federalism 42, 234–259.

Hepp, R., Von Hagen, J., 2013. Interstate risk sharing in Germany: 1970-2006. Oxford Economic Papers 65, 1–24.

Husted, L., Rogers, J., Sun, B., 2020. Monetary policy uncertainty. Journal of Monetary Economics 115, 20–36.

Jurado, K., Ludvigson, S.C., Ng, S., 2015. Measuring uncertainty. American Economic Review 105, 1177–1216.

Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. Econometrica 47, 263–292.

Kapetanios, G., 2008. A bootstrap procedure for panel data sets with many cross sectional units. Econometrics Journal 11, 377–395.

Kenen, P., 1969. The theory of optimum currency areas: an eclectic view. In *Monetary Problems of the International Economy*, edited by Robert A. Mundell and Alexander K. Swoboda. Chicago: Chicago University Press .

Lettau, M., Maggiori, M., Weber, M., 2014. Conditional risk premia in currency markets and other asset classes. Journal of Financial Economics 114, 197–225.

Mace, B.J., 1991. Full insurance in the presence of aggregate uncertainty. Journal of Political Economy 99, 928–956.

Malkin, I., Wilson, D.J., 2013. Taxes, transfers, and State economic differences. Economic Letter, Federal Reserve Bank of San Francisco 36.

Markowitz, H., 1959. Portfolio selection: efficient diversification of investments. Yale University Press, New Haven and London .

Mélitz, J., Zumer, F., 1999. Interregional and international risk-sharing and lessons for EMU. Carnegie-Rochester Conference Series on Public Policy 51, 149–188.

Milano, V., 2017. Risk sharing in the eurozone: the role of European institutions. CeLEG Working Papers Series - LUISS Guido Carli .

Nelson, J.A., 1994. On testing for full insurance using Consumer Expenditure Survey data. Journal of Political Economy 102, 384–394.

Obstfeld, M., Peri, G., 1998. Regional non-adjustment and regional shocks. Economic Policy 13, 205–259.

Plagborg-Møller, M., Reichlin, L., Ricco, G., Hasenzagl, T., 2020. When is growth at risk? Brookings Papers on Economic Activity , 167–229.

Poghosyan, T., Senhadji, A., Cottarelli, C., 2016. The role of fiscal transfers in smoothing regional shocks: evidence from existing federations. IMF Working Papers .

Routledge, B., Zin, S., 2010. Generalized disappointment aversion and asset prices. The Journal of Finance 65, 1303–1332.

Roy, A., 1952. Safety first and the holding of assets. Econometrica 20, 431–449.

Sala-i-Martin, X., Sachs, J., 1992. Fiscal policies and optimum currency areas: evidence for Europe from the United States. In Matthew Canzoneri, Vittorio Grilli, and Paul Masson (eds.), Establishing a Central Bank: Issues in Europe and Lessons from the United States. Cambridge: Cambridge University Press , 195–219.

Simon, N., Friedman, J., Hastie, J., Tibshirani, R., 2013. A sparse-group Lasso. Journal of Computational and Graphical Statistics 22, 231–245.

Sørensen, B.E., Yosha, O., 1998. International risk sharing and European monetary unification. Journal of International Economics 45, 211–238.

Stempel, D., 2021. Risk sharing heterogeneity in the United States. Economics Bulletin, Heinrich Heine University Düsseldorf 41, 1223–1240.

Tax Foundation, 2025. Historical US federal individual income tax rates and brackets, 1862-2025 .

US Department of Health and Human Services, 2021. Welfare indicators and risk factors. 20th Report to Congress .

Von Hagen, J., 1992. Fiscal arrangements in a monetary union: evidence from the US. In D. Fair and C. de Boissieux (eds), Fiscal Policy, Taxes, and the Financial System in an Increasingly Integrated Europe, Kluwer , 337–359.